Matplotlib Graphs

Assignment No. 3 | Made by: Raza Mustafa

Introduction

Matplotlib is a Python library used to draw graphs and charts from data. We mostly use its `pyplot` module.

```
import matplotlib.pyplot as plt
```

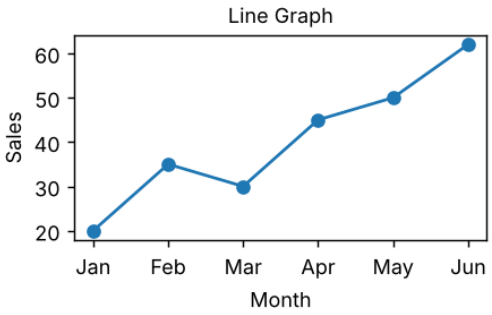
Every graph ends with `plt.show()` to display it. `plt.title()`, `plt.xlabel()` and `plt.ylabel()` add a title and axis labels.

Types of Graphs

1. Line Graph

Shows how a value changes over time, e.g. monthly sales.

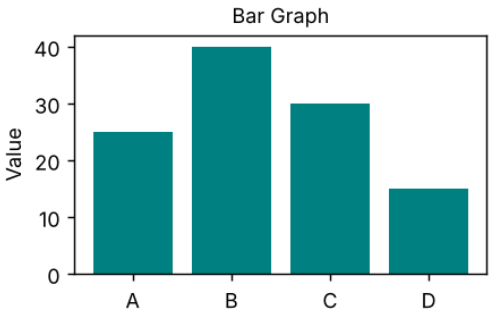
```
months = ['Jan','Feb','Mar','Apr','May','Jun']
sales = [20, 35, 30, 45, 50, 62]
plt.plot(months, sales, marker='o')
plt.show()
```



2. Bar Graph

Compares values of different categories using vertical bars.

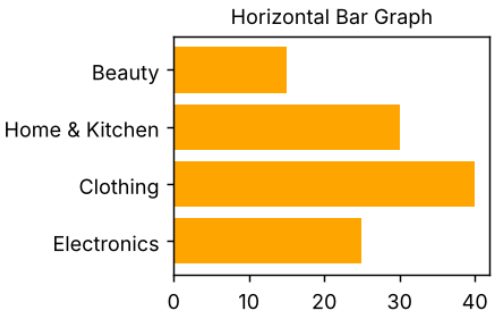
```
cats = ['A','B','C','D']
vals = [25, 40, 30, 15]
plt.bar(cats, vals)
plt.show()
```



3. Horizontal Bar Graph

Same as a bar graph but bars are horizontal. Useful when category names are long.

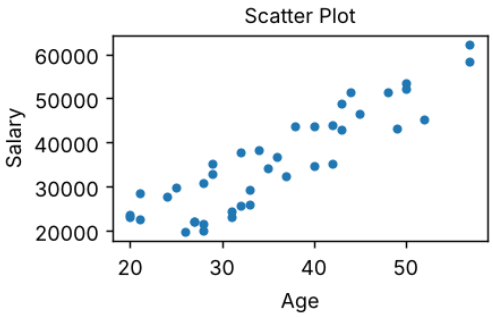
```
plt.barh(['Electronics','Clothing',
          'Home & Kitchen','Beauty'], vals)
plt.show()
```



4. Scatter Plot

Shows the relationship between two numeric variables using dots, e.g. age vs salary.

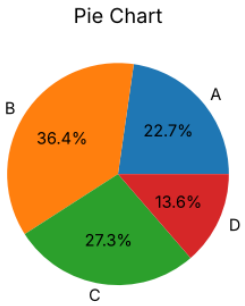
```
plt.scatter(age, salary)
plt.xlabel('Age')
plt.ylabel('Salary')
plt.show()
```



5. Pie Chart

Shows parts of a whole as percentages.

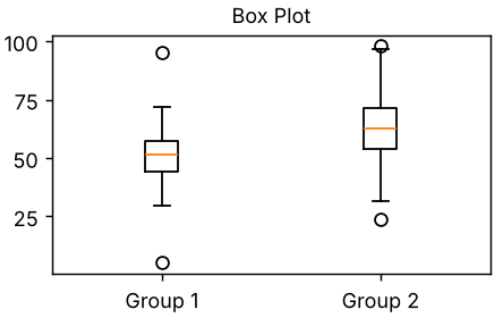
```
plt.pie(vals, labels=cats,
        autopct='%1.1f%%')
plt.show()
```



6. Box Plot

Shows the spread of data (min, Q1, median, Q3, max). Dots outside are outliers.

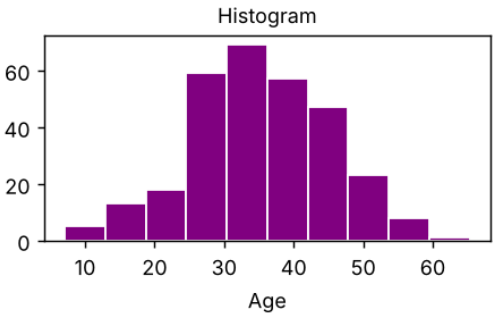
```
plt.boxplot([group1, group2])
plt.show()
```



7. Histogram

Shows how often values fall in different ranges (bins), e.g. age distribution.

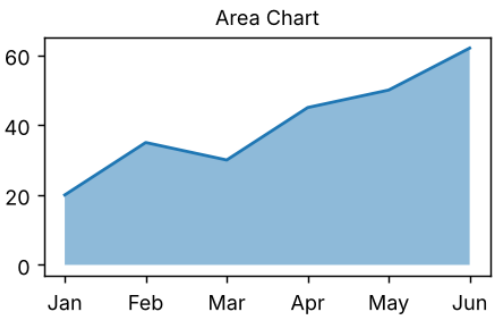
```
plt.hist(ages, bins=10)
plt.show()
```



8. Area Chart

A line graph with the area below it filled. Shows total amount over time.

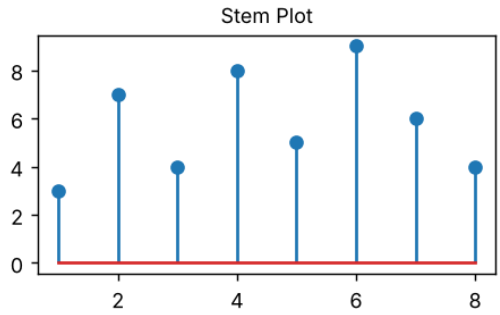
```
plt.fill_between(range(6), sales,
                 alpha=0.5)
plt.show()
```



9. Stem Plot

Shows each value as a vertical line with a dot on top. Good for small datasets.

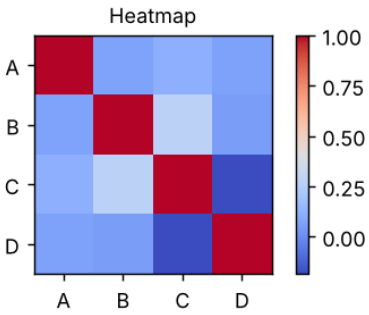
```
plt.stem([1,2,3,4,5,6,7,8],
         [3,7,4,8,5,9,6,4])
plt.show()
```



10. Heatmap

Shows values of a 2D table using colors. Often used for a correlation matrix.

```
plt.imshow(matrix, cmap='coolwarm')
plt.colorbar()
plt.show()
```



Summary

Graph	Function	Best used for
Line	`plt.plot()`	Trend over time
Bar	`plt.bar()`	Comparing categories
Horizontal Bar	`plt.barh()`	Categories with long names
Scatter	`plt.scatter()`	Relationship between two variables
Pie	`plt.pie()`	Parts of a whole
Box	`plt.boxplot()`	Spread of data and outliers
Histogram	`plt.hist()`	Distribution of data
Area	`plt.fill_between()`	Total amount over time
Stem	`plt.stem()`	Individual values in small data
Heatmap	`plt.imshow()`	2D data / correlation